MAgent Theory 2.0

Formation with Top-Down Feedback:
Shared Labels and Exact Parents

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Status and summary

This is a working document for the seventh item of the future-work list, consequences for formation. It takes up the open problem on formation with top-down feedback left by the Item 3 working document. It is not part of the main manuscript and is not cited by it. The question is whether parents created by a state-determined handoff when a wall forms, and then acting back on their members, give an inherited recursive formation in the sense of the standard in chapter 5 of the main manuscript, clauses (F1) to (F6), with the null flow as the control for interaction necessity. The constructions are stated for general probability laws. The Gaussian location family is the computational instance, and every statement that depends on it says so.

Feedback of the obvious kinds does not change what forms. A parent added as one more source in its members' rows multiplies the row part of each member's force by a common factor when the member agrees with it, leaves record forces unchanged, and alters no exponent of the rows (proposition 2.1); in an example without records the same blocks form and only the second stage is delayed. In an objective with a gradient flow, such as the global joint of Items 2 and 3 or the reciprocal potential with fixed rows, a top-down cost invariant under the collective motion of a block exerts no force on the block's momentum whenever that motion has a momentum function, as it does in the Gaussian location family (proposition 2.2). An exactly inherited pooled parent therefore does not see it; the sharpening of walls reported in Item 3 passes through the distribution of that momentum among the members, which sparse bridges read. A post-handoff flow with a globally attracting equilibrium cannot meet the counterfactual clause (F6b) (proposition 2.3), and the Gaussian tangent global joint of Items 2 and 3 is strictly convex, so no declared Gaussian hierarchy in it can.

What changes outcomes is the pooling of evidence. When a block forms, its members share one source label: they infer jointly which source to trust, with the prior over sources applied once and their costs added. For general laws the shared row depends on the block only through the linear pool of the members' laws, and it equals the Gibbs row of a single receiver holding that pool at temperature T/n (proposition 3.2); for sources in an exponential family it depends only on the pooled mean parameter. The parent presents the pool to its members and to every receiver outside the block. A tempered label, which averages the members' costs instead of adding them, reproduces the dynamics without feedback on internally synchronized configurations (proposition 3.4), so the factor n in the exponent is the operative part of the feedback.

In the Gaussian location family the shared-label parent is exactly inherited and compresses. The pooled mean of every block evolves autonomously, by the per-agent law of a single agent at temperature T/n with one additional source at its own law, while the members' deviations from the pool relax at a fixed rate to offsets determined by their records, whatever the spread within the block (theorem 4.1). External equitable partitions give exact block-mean dynamics for linear consensus on a fixed graph [15]; here the rows are state-dependent Gibbs rows. The result does not extend beyond this family: if the pooled parent of a two-member block closes for every configuration and

every target, the Fisher metric is constant and the family is a Gaussian location family (theorem 4.3). For Bernoulli laws it fails.

In a twelve-agent instance with four groups of three conflicting records, the null flow never selects a block (proposition 5.1). The shared-label flow selects the four groups and then, on each side, the union of two first-stage parents. The second stage reduces to a one-dimensional equation for the gap between two parents, with a stable merged state, an unstable middle root, and a stable separated state held by the records. The union forms exactly when the first stage leaves the parents inside the middle root (proposition 5.3 and corollary 5.4). Counterfactual handoffs outside it never unite, while the control without feedback unites from every tested handoff. At slightly larger record gaps the hierarchy stops at depth one in the separated state, and the distance from which two blocks still unite shrinks with block size roughly as $n^{-1/2}$.

Measured against the standard, the instance meets (F1) to (F6) (chapter 6). The qualifications are these. The selector thresholds and the label-sharing handoff are declared, and the compression interface is derived from that rule, not from dynamics that are unchanged at the handoff. The compressed coordinates are transient and converge to record-determined offsets. Exactness holds only in the Gaussian location family. The first-stage events and the unions from the actual handoffs come from fixed-step numerical integration with explicit margins, not from validated arithmetic. The strong form of (F5) fails, and the parameter window is narrow. For general laws the pooled parent is not exactly inherited, and the inheritance clause of the general-law construction is open. A bounded literature search found exact quotients of linear consensus, Bayesian inference of latent social groups, representative agents for coherent groups, and Bayesian structure learning of hierarchies, but no construction combining shared-label pooling, exact block-mean closure away from synchrony, and merging that depends on the state at formation (chapter 8). This is not a priority claim.

A derived selector replaces the declared thresholds by a comparison of free energies (chapter 7). A block is selected when the shared-label models that the handoff would create, together with the models of all other units, have a lower free energy than the current ones, and the comparison has held for the declared window. For general laws this structure free energy reads only the spread of each block, n times a generalized Jensen–Shannon divergence, and the free energy of its pool at temperature T/n, which includes the block's own parent at cost zero (proposition 7.2). Two agents whose mutual attentions have a product above the peer prior gain from a shared label, through a saving of labels and a double use of their laws in the pool (proposition 7.3). The derived selector therefore groups whatever attention the initial state contains, and in the instance the null flow selects the four groups at once. Interaction necessity moves to the second stage. The criterion for uniting two parents is a radius, $d_* = 0.7685$ for blocks of three, below the middle root of the gap equation (proposition 7.6), so the counterfactual dependence survives (corollary 7.7). With $\tau_p = 1.25$ both flows select the four groups and only the shared-label flow unites them. The window now decides the regime: with an earlier handoff the hierarchy stops at depth one, as the null does, and with the window $\tau_p = 2$ of the declared instance the flow selects the two unions directly as a single stage. No window gives interaction necessity at both stages in this instance.

Every identity was checked numerically at small sizes in `item7_checks/checks.py`, whose output is `checks_output.txt`; the checks support the instances tested and are not proofs. Statements that rest on numerical integration are labeled as such.

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Chapter 1

Setting

The agents $i \in \mathcal{V}$, $|\mathcal{V}| = N$, hold belief laws q_i on a standard Borel space Y. The model sector is trivial in this document; it enters only through a common evaluator G, a law on Y shared by agents of one species. In the per-agent mixture-source model of chapter 3 of the main manuscript, agent i has a finite source set $\mathcal{N}_i$ containing its own label, with positive prior row π_i. Its self cost is $D_{ii} = D_{\mathrm{KL}}(q_i \| G)$, its cost for a peer j is $D_{ij} = D_{\mathrm{KL}}(q_i \| \nu_{ij})$ with ν_{ij} the law of j as compared by i, and its records o_i enter through a likelihood $\ell_i(o_i \mid k)$ that persists in the objective. At temperature T the per-agent free energy is

$$\mathcal{F}_i = \sum_{\nu \in \mathcal{N}_i} \beta_{i\nu} D_{i\nu} + T D_{\mathrm{KL}}(\beta_i \| \pi_i) + \int -\log \ell_i(o_i \mid k) q_i(dk), \tag{1.1}$$

and its minimizing row is the Gibbs row $\beta_{i\nu} = \pi_{i\nu} e^{-D_{i\nu}/T} / Z_i$. The main manuscript's per-agent model has $T = 1$; another temperature corresponds to tempered factors. Each agent follows the Fisher gradient flow of its own $\mathcal{F}_i$, with the laws of the other agents held as data. By the envelope identity, the differential of $\mathcal{F}_i$ at its Gibbs row is $\sum_\nu \beta_{i\nu} dD_{i\nu}$ plus the differential of the record term, so the force on q_i is a row-weighted sum of divergence forces plus a record force that no row weights.

Definition 1.1 (Units, presented laws and the wall selector). At each time the agents are partitioned into units: the blocks selected so far that are contained in no later block, and the agents that belong to no block. Each unit U presents a law P_U, which is the agent's own law for an agent and is declared by the handoff of definition 3.1 for a block. Fix thresholds $0 < \delta_c < \delta_w$ and a window $\tau_p > 0$. Units $U \neq V$ are coherent if $D_{\mathrm{KL}}(P_U \| P_V) < \delta_c$ and $D_{\mathrm{KL}}(P_V \| P_U) < \delta_c$. A set C of at least two units is a candidate if it is a connected component of the coherence graph and if $D_{\mathrm{KL}}(P_U \| P_V) > \delta_w$ and $D_{\mathrm{KL}}(P_V \| P_U) > \delta_w$ for every $U \in C$ and every unit $V \notin C$. A handoff selects the union of the agents of C as a block once C has been a candidate without interruption for a time τ_p. A selected block is never dissolved.

The selector reads only the current presented laws and their recent history, so each handoff is a measurable function of accessible history, as (F2) requires. It is a declared threshold selector of the kind used in chapters 11 to 13 of the main manuscript. Its block relation, internal coherence together with a wall, is the contrast structure that Item 3 found at stationary states with parents.

Definition 1.2 (Controls). The null flow Φ^0 is that of chapter 5 of the main manuscript. Every cost D_{ij} with $j \neq i$ is deleted, while labels and priors are kept, so the null self row is

$$\beta^0_{ii} = \frac{\pi_{ii} e^{-D_{ii}/T}}{\pi_{ii} e^{-D_{ii}/T} + 1 - \pi_{ii}}, \tag{1.2}$$

the convention of chapter 13 of the main manuscript. The control without feedback Φ^{off} uses the same selector and records the same handoffs, but every agent keeps its individual row, sources and channels after a handoff, so the parents are passive.

The control without feedback is not part of the standard. It answers a different question: whether the action of a parent on its members changes what forms.

Chapter 2

What top-down feedback cannot do

Proposition 2.1 (A parent as one more source). Let receiver i have the Gibbs row β_i over a finite source set $\mathcal{N}_i$ at temperature T, and add a source A with prior share $a \in (0,1)$, multiplying the other priors by $1-a$. The new row satisfies

$$\beta'_{i\nu} = (1-\beta'_{iA})\beta_{i\nu}, \qquad \nu \in \mathcal{N}_i. \tag{2.1}$$

If the first variation of D_{iA} vanishes at the current law of i, as it does when the compared law of A equals q_i, then the force on q_i is $1-\beta'_{iA}$ times its row part before the addition, plus the unchanged record force.

Proof. With $Z'_i = (1-a)Z_i + ae^{-D_{iA}/T}$ we have $\beta'_{i\nu} = (1-a)\pi_{i\nu}e^{-D_{i\nu}/T}/Z'_i = \beta_{i\nu}(1-a)Z_i/Z'_i$ and $1-\beta'_{iA} = (1-a)Z_i/Z'_i$, which is eq. (2.1). By the envelope identity the row part of $d\mathcal{F}'_i$ is $\sum_\nu \beta'_{i\nu}dD_{i\nu} + \beta'_{iA}dD_{iA}$, which equals $(1-\beta'_{iA})\sum_\nu \beta_{i\nu}dD_{i\nu}$ where $dD_{iA}=0$. The record term is weighted by no row, and the Fisher field is linear in the differential. □

The exponents of the row are unchanged, so the parent does not change which of the old sources the member trusts. Without records, the parent only rescales each member's velocity by a positive factor at agreement. With records, it raises the weight of the records relative to the row part by the factor $(1-\beta'_{iA})^{-1}$, a form of stubbornness. Away from agreement it also pulls the member toward its compared law, which reduces the spread inside the block. Check (2) confirms eq. (2.1) and the force decomposition. Check (15) runs twelve agents in four groups without records, each agent's evaluator target in its own row and each member adding its pooled parent with share a; for $a = 0$, 0.3 and 0.6 the same four blocks and the same two unions are selected, and the unions occur at $t \approx 40$, 100 and 245.

Proposition 2.2 (Top-down costs are internal forces). Let $\mathcal{E} = \mathcal{E}_0 + \tau\mathcal{E}_{\mathrm{td}}$ be continuously differentiable on a Riemannian manifold (S, g), and let $x(t)$ solve $\dot{x} = -\operatorname{grad}_g \mathcal{E}$. Let J be continuously differentiable on S with $\xi = \operatorname{grad}_g J$. If $\mathcal{E}_{\mathrm{td}}$ is invariant along ξ, that is $d\mathcal{E}_{\mathrm{td}}(\xi) = 0$, then

$$\frac{d}{dt}J(x(t)) = -d\mathcal{E}_0(\xi)\big(x(t)\big), \tag{2.2}$$

which does not involve τ.

Proof. $\frac{d}{dt}J(x) = dJ(\dot{x}) = g(\xi,\dot{x}) = -g(\xi, \operatorname{grad}_g \mathcal{E}) = -d\mathcal{E}(\xi) = -d\mathcal{E}_0(\xi)$. □

In the Gaussian location family with laws $\mathcal{N}(y_i, P_i^{-1})$, the Fisher metric on the means is $\bigoplus_i P_i$. The common translation of the members of a block I and of a parent A by $u \in \mathbb{R}^K$ is $\xi_u = \operatorname{grad} J_u$ with $J_u = u^T(\sum_{i\in I} P_i y_i + P_A y_A)$. The top-down factor of Item 3 depends on $y_i - y_A$ and on the

cross-scale dictionary, so its cost is invariant under ξ_u, and the block momentum $\sum_{i\in I} P_i y_i + P_A y_A$ has a rate independent of τ at every configuration (check (3a)). Suppose in addition that the parent is the pool, with top-down cost $\frac{\tau}{2}\sum_{i\in I}\|y_i - \bar{y}_I\|^2$, that the members have equal precisions, and that every member of a block attends to every member of every other block with one weight. Then the block sums form a closed linear system, the external equitable partition of O'Clery et al. [15], and their trajectories do not depend on τ (check (3b), residual 6×10^{-16}). With sparse bridges, or with a parent that has its own state and prior, the block sums do depend on τ (checks (3b) and (3c)). The top-down cost moves momentum between the parent and the members and among the members, and the bridges read where it sits. The sharpening of walls under top-down coupling found in Item 3 is of this kind, and it is not available to a parent that is exactly inherited through the block momentum.

For general laws the proposition needs a momentum function for the collective motion. In natural coordinates of an exponential family the common shift of natural parameters has momentum $\sum_i \mu_i$, the sum of mean parameters. A Kullback–Leibler cost between members of the family is a Bregman divergence of the log-partition ψ, and differentiating it along a common shift shows that it is invariant for every pair exactly when $\nabla\psi$ is affine. Outside the Gaussian location family the proposition therefore applies to costs constructed to be invariant, not to the divergences of the theory.

Proposition 2.3 (A globally attracting equilibrium excludes counterfactual recursion). Fix the first-stage blocks of a construction and let $\mathcal{C}$ be a set of post-handoff states, the counterfactual handoffs of (F6b). Suppose the post-handoff flow is deterministic and every trajectory starting in $\mathcal{C}$ converges to one equilibrium z^*. Let the relation that some second-stage block B is a candidate be given by strict inequalities between continuous functions of the state, defining an open set S, and let persistence mean that the relation holds at all later times. If $z^* \in S$, the block B is selected from every state in $\mathcal{C}$ and (F6b) fails for B. If $z^* \notin \bar{S}$, no state in $\mathcal{C}$ leads to a persistent selection of B.

Proof. If $z^* \in S$, some neighborhood of z^* lies in S, and every trajectory from $\mathcal{C}$ remains in it after a finite time, so the candidate persists beyond any window τ_p. If the relation holds at all times after some t_0 along a trajectory converging to z^*, then z^* is a limit of points of S. □

The Gaussian tangent global joint of Items 2 and 3 has a positive definite quadratic energy in the contents and the dictionaries whenever the content anchors and the dictionary priors are positive definite. Declared Gaussian parents add positive semidefinite cross-scale terms and a parent anchor. The gradient flow of this energy in any constant metric therefore converges to its unique minimizer from every state, every counterfactual handoff with the same first-stage blocks leads to the same limit, and (F6b) fails for every persistent second stage that is selected at all. Counterfactual recursion needs several stable states after a handoff, hence a nonconvex objective: Gibbs rows, dictionaries acting through the exponential map, or non-Gaussian laws. The construction below uses Gibbs rows.

Chapter 3

Shared labels

Definition 3.1 (Shared-label handoff). Let a handoff select a block I of $n = |I| \geq 2$ agents. Suppose that every member compares each source outside I through the same channel, so that a source ν presents one compared law ν_ν to all members, and that the members share the evaluator G. From the handoff on:

(a) The members carry one source label J_I. Its sources are the evaluator, a parent A_I, and the units outside I. Its prior π_I is the average of the members' priors, with the mass that the members gave to one another assigned to A_I and the mass that they gave to the agents of an outside unit assigned to that unit. Each member descends, with respect to its own law and with all other laws as data, the block free energy

$$\mathcal{F}_I = \sum_\nu \beta_{I\nu} \sum_{i\in I} D_{i\nu} + T D_{\mathrm{KL}}(\beta_I \| \pi_I) + \sum_{i\in I} \int -\log \ell_i(o_i \mid k) q_i(dk), \tag{3.1}$$

whose minimizing row is $\beta_{I\nu} = \pi_{I\nu} \exp(-\sum_{i\in I} D_{i\nu}/T)/Z_I$.

(b) The parent presents the linear pool $\bar{q}_I = n^{-1} \sum_{i\in I} q_i$ or, when the compared laws lie in a declared family E, its moment projection P_I onto E. The members compare themselves to it, $D_{iA} = D_{\mathrm{KL}}(q_i \| P_I)$, and every receiver outside I replaces the agents of I in its source set by A_I, with their total prior.

(c) A later handoff that selects a set of units replaces them in the same way by one block whose members are all their agents. The parents of the replaced blocks are then no longer sources.

The tempered label replaces T by nT in the row of (a).

Part (a) is top-down attention: a member no longer decides alone which source to trust; the block decides, with every member's evidence. Part (b) is top-down prediction and routing: members are pulled toward what the block presents, and the block speaks to the rest of the population through its parent. The rule is declared once, before any selection, and applies at every level, so a second stage uses it without a new parameter.

Proposition 3.2 (The shared row reads the pool). Under definition 3.1, for every source ν whose costs are finite,

$$\sum_{i\in I} D_{\mathrm{KL}}(q_i \| \nu_\nu) = \sum_{i\in I} D_{\mathrm{KL}}(q_i \| \bar{q}_I) + n D_{\mathrm{KL}}(\bar{q}_I \| \nu_\nu). \tag{3.2}$$

Hence

$$\beta_{I\nu} = \frac{\pi_{I\nu} \exp\big(-n D_{\mathrm{KL}}(\bar{q}_I \| \nu_\nu)/T\big)}{\sum_\mu \pi_{I\mu} \exp\big(-n D_{\mathrm{KL}}(\bar{q}_I \| \nu_\mu)/T\big)}, \tag{3.3}$$

the Gibbs row at temperature T/n of a single receiver with law $\bar{q}_I$, and the parent presenting $\bar{q}_I$ has relative cost zero. If every compared law lies in a regular exponential family E with sufficient statistic t, and the parent presents the moment projection P_I, defined by $\mathbb{E}_{P_I} t = \bar{\mu}_I = n^{-1} \sum_i \mathbb{E}_{q_i} t$, then for every $p \in \mathsf{E}$

$$D_{\mathrm{KL}}(\bar{q}_I \| p) = D_{\mathrm{KL}}(\bar{q}_I \| P_I) + D_{\mathrm{KL}}(P_I \| p), \tag{3.4}$$

and the shared row is the Gibbs row at temperature T/n of the costs $D_{\mathrm{KL}}(P_I \| \nu_\nu)$. It depends on the block only through $\bar{\mu}_I$, and A_I again has relative cost zero.

Proof. Write $\log(q_i/\nu) = \log(q_i/\bar{q}_I) + \log(\bar{q}_I/\nu)$, integrate against q_i, and sum over i, using $\sum_i q_i = n\bar{q}_I$; this is the receiver-pool identity of chapter 3 of the main manuscript with equal weights. The term $\sum_i D_{\mathrm{KL}}(q_i \| \bar{q}_I)$ is the same for every source and cancels from the normalized row, and with the parent presenting $\bar{q}_I$ it is the whole cost of A_I. For eq. (3.4), write $p = p_\theta$ with log density $\theta \cdot t - \psi(\theta)$ relative to a base measure; then $D_{\mathrm{KL}}(\bar{q}_I \| p_\theta) = -H(\bar{q}_I) - \theta \cdot \bar{\mu}_I + \psi(\theta)$ with H the entropy relative to that measure. Subtracting the same expression at θ^* with $\nabla\psi(\theta^*) = \bar{\mu}_I$ gives $\psi(\theta) - \psi(\theta^*) - (\theta - \theta^*) \cdot \nabla\psi(\theta^*) = D_{\mathrm{KL}}(p_{\theta^*} \| p_\theta)$, the Pythagorean identity of Csiszár [2]. The cost of A_I is then the common term plus $nD_{\mathrm{KL}}(\bar{q}_I \| P_I)$, which is also common to all sources. □

Check (1) confirms eqs. (3.2) to (3.4) on a six-point alphabet, including that two blocks with the same pool and different members have the same row.

Remark 3.3 (When the shared row is a posterior). In the registered-agreement model of chapter 3 of the main manuscript, an agent's Gibbs row is the posterior of the source it heard, given that it registered agreement with that source, at temperature equal to the reciprocal message count. If the members of I heard one common source J_I and their registrations are conditionally independent given J_I, the posterior of J_I is π_I times the product of the members' registration probabilities, and the source-independent factors cancel; it is the row of definition 3.1(a). The shared row is then the exact posterior of a common cause. Members that have influenced one another do not in general supply conditionally independent evidence, and the row is then a Gibbs row of summed losses, a generalized-Bayes update in the sense of Bissiri et al. [1], not a posterior. It adds the members' costs, unlike a logarithmic opinion pool with weights summing to one [5]; the tempered label is the pool that averages them.

Proposition 3.4 (A tempered label is neutral at synchrony). Suppose the priors are homogeneous, $\pi_{ii} = \pi_s$ and $\pi_{ij} = \pi_p$ for $j \neq i$, and all channels are identity channels. At a configuration where every block is internally synchronized, $q_i = \bar{q}_I$ for $i \in I$, the force on every agent under the shared-label handoff with the tempered label equals its force under the control without feedback. With the untempered label, every cost in the row of a block of n members is n times the corresponding cost in the control.

Proof. At synchrony, in the control, member i of I has the evaluator cost $D_{\mathrm{KL}}(\bar{q}_I \| G)$, cost zero to each of the other $n-1$ members with total prior $(n-1)\pi_p$, and cost $D_{\mathrm{KL}}(\bar{q}_I \| \bar{q}_V)$ to each of the n_V agents of an outside unit V, with total prior $n_V\pi_p$; the sources at cost zero exert no force. The tempered shared row has costs $n^{-1} \sum_{i \in I} D_{i\nu}$, which at synchrony are $D_{\mathrm{KL}}(\bar{q}_I \| G)$, zero for A_I with prior $(n-1)\pi_p$, and $D_{\mathrm{KL}}(\bar{q}_I \| P_V) = D_{\mathrm{KL}}(\bar{q}_I \| \bar{q}_V)$ for V with prior $n_V\pi_p$, because $P_V = \bar{q}_V$ at

synchrony. The rows, the compared laws and the record forces coincide. Without tempering each cost is the sum of n equal member costs. □

Check (6) gives a difference of 2×10^{-16} between the tempered label and the control at a synchronized configuration, and 0.06 for the untempered label. At synchrony the whole effect of the shared label is evidence accumulation: a block of n members is n times as selective as one of its members. The tempered label differs from the control only through the spread inside blocks.

Remark 3.5 (Dictionaries). With receiver-owned dictionaries, as in Items 1 to 3, member i reads source ν through its own dictionary $\bar{\theta}_{i\nu}$, acting by push-forward, and the compared laws differ across members. Relative entropy is invariant under one measurable bijection applied to both arguments, so if there are frame changes g_i of the members with $g_i\bar{\theta}_{i\nu} = \Theta_\nu$ for every source ν outside I, then $\sum_i D_{\mathrm{KL}}(q_i\|\bar{\theta}_{i\nu}\nu) = \sum_i D_{\mathrm{KL}}(g_iq_i\|\Theta_\nu\nu)$, and proposition 3.2 applies to the laws g_iq_i. Such frame changes exist exactly when $\bar{\theta}_{j\nu}\bar{\theta}_{i\nu}^{-1}$ does not depend on ν for all members i, j: then $g_j = \bar{\theta}_{1\nu}\bar{\theta}_{j\nu}^{-1}$ works for every ν, and conversely $g_j^{-1}g_i = \bar{\theta}_{j\nu}\bar{\theta}_{i\nu}^{-1}$. Equivalently, every loop through two members and two outside sources has trivial holonomy. In the language of Item 3 the block is flat toward its sources, and Θ_ν is the meta-level dictionary of its flat-wall proposition. For a block that is curved toward its sources the shared row is not a function of the pool of any frame-changed members.

Chapter 4

Exact parents in the Gaussian location family

Theorem 4.1 (Exact inheritance of shared-label parents). Let $\mathsf{Y} = \mathbb{R}^K$, let every agent law, compared law, presented law and the evaluator be Gaussian with one fixed precision matrix Λ, recognition being restricted to the location family $\mathcal{N}(y, \Lambda^{-1})$ whose Fisher metric on the means is Λ, let all channels be identity channels, and let each record have the likelihood $\mathcal{N}(o_i; k, R^{-1})$; write $\rho = \Lambda^{-1}R$. The parent of a block presents $\mathcal{N}(\bar{y}_U, \Lambda^{-1})$, the moment projection of its pool onto the location family. After any sequence of shared-label handoffs, let $\mathcal{U}$ be the set of units, n_U their sizes, $\bar{y}_U$ and $\bar{o}_U$ the averages of the members' means and records, and m_ν the mean of source ν: m_G for the evaluator, $\bar{y}_U$ for the unit's own parent, and $\bar{y}_V$ for a unit $V \neq U$. Then for every $U \in \mathcal{U}$ and $i \in U$:

(i) $\beta_{U\nu} = \pi_{U\nu} \exp\!\big(-\frac{n_U}{2T}\|\bar{y}_U - m_\nu\|^2_\Lambda\big)/Z_U$;

(ii) $\dot{y}_i = \sum_\nu \beta_{U\nu}(m_\nu - y_i) + \rho(o_i - y_i)$;

(iii) $\frac{d}{dt}\bar{y}_U = \sum_\nu \beta_{U\nu}(m_\nu - \bar{y}_U) + \rho(\bar{o}_U - \bar{y}_U)$;

(iv) $\frac{d}{dt}(y_i - \bar{y}_U) = -(I + \rho)(y_i - \bar{y}_U) + \rho(o_i - \bar{o}_U)$.

Consequently the retention $R(x) = (\bar{y}_U)_{U\in\mathcal{U}}$ satisfies, on every set of post-handoff states, the dynamical condition (D), the response condition (B) for every protocol in which a receiver outside a block attends to its parent, and the channel condition (C) of the definition of an inherited parent in chapter 5 of the main manuscript, with parent flow (iii). The retention is not injective when some unit has $n_U \geq 2$, and two configurations in one fiber remain distinct and in one fiber at all later times.

Proof. (i) For a source with mean m,

$$\sum_{i\in U} D_{\mathrm{KL}}\big(\mathcal{N}(y_i, \Lambda^{-1}) \| \mathcal{N}(m, \Lambda^{-1})\big) = \frac{n_U}{2}\|\bar{y}_U - m\|^2_\Lambda + \frac{1}{2}\sum_{i\in U}\|y_i - \bar{y}_U\|^2_\Lambda, \tag{4.1}$$

and the second term, which is the whole cost of the own parent, does not depend on the source; it cancels from the normalized row. (ii) The derivative of eq. (3.1) with respect to y_i is $\sum_\nu \beta_{U\nu}\Lambda(y_i - m_\nu) + R(y_i - o_i)$, since the record term is $\frac{1}{2}\|o_i - y_i\|^2_R$ plus a constant. The Fisher flow multiplies by $-\Lambda^{-1}$, and the rows sum to one. (iii) Average (ii) over $i \in U$; the rows and the source means do not depend on i. (iv) Subtract (iii) from (ii) and use $\sum_\nu \beta_{U\nu} = 1$.

By (i) the rows of every unit are functions of the unit means. By the routing of definition 3.1(b) every compared law is the evaluator or a presented law $\mathcal{N}(\bar{y}_V, \Lambda^{-1})$, so (iii) is a closed system in $(\bar{y}_U)_U$ and $R \circ \Phi_t = \Phi^c_t \circ R$ for the flow Φ^c of (iii). A receiver outside U that attends to A_U receives $\mathcal{N}(\bar{y}_U, \Lambda^{-1})$, a function of R; the rows, channels and evaluator read by (iii) are functions of R. The

deviations $(y_i - \bar{y}_U)_{i\in U}$ with zero sum span a subspace of dimension $(n_U - 1)K$ on which R is constant. By (iv) they evolve by an invertible affine flow that does not depend on the unit means, so distinct deviations stay distinct while the unit means move together by (iii). □

Check (7) integrates the twelve-agent instance of chapter 5 for twenty time units after its first-stage handoffs. The block means agree with the solution of (iii) to 8×10^{-15}, and the deviations agree with the closed form of (iv) to 1.4×10^{-15}. The external equitable partitions of linear consensus give exact block-mean dynamics on a fixed graph [15], and for nonlinear networks the quotient over such partitions describes the synchronized states [18]. Here the rows depend on the state, the closure holds for every spread inside the blocks, and it is created by the handoff rule.

Corollary 4.2 (Composition and type). The parent flow (iii) is the per-agent flow of a single Gaussian location agent with mean $\bar{y}_U$ and record $\bar{o}_U$, at temperature T/n_U, whose source set contains, besides the evaluator and the other units, a source at its own law with prior π_{UA}. The class of such agents, with any temperature and any prior on a source at the agent's own law, is closed under shared-label handoffs, and successive retentions compose by the composition proposition of chapter 5 of the main manuscript. A block formed from blocks I and J has temperature $T/(n_I+n_J)$, and the retention from the two parents to their union, $(\bar{y}_I, \bar{y}_J) \mapsto (n_I\bar{y}_I + n_J\bar{y}_J)/(n_I + n_J)$, is again not injective.

Proof. The first statement is (iii) read as the flow (ii) of a single agent with the stated rows; a single agent is the case $n_U = 1$, with prior zero on the source at its own law. A handoff applied to units of this class adds their costs, by (i), and returns a unit of the class with the summed size. Composition is the composition of the two exact retentions. □

Theorem 4.3 (Exact pooled parents characterize the Gaussian location family). Let the members' laws and all compared laws lie in a regular exponential family with sufficient statistic $t : \mathsf{Y} \to \mathbb{R}^d$, natural parameter θ in an open set Θ, log-partition ψ, and mean parameter $\mu = \nabla\psi(\theta)$. Let the dynamics be the per-agent Fisher flow of the block free energy eq. (3.1) without records, with the parent presenting the moment projection of the pool. Suppose that for blocks of two members the velocity of the pooled mean parameter $\bar{\mu} = (\mu_1 + \mu_2)/2$ is, at every configuration, a function of $\bar{\mu}$ and of the row-weighted source parameter $c = \sum_\nu \beta_\nu\theta_\nu$, for every value of c in Θ. Then $\nabla^2\psi$ is constant on Θ, and the laws of t form a Gaussian family with one covariance and varying mean; when t is the identity on $\mathsf{Y} = \mathbb{R}^d$, the family is the Gaussian location family.

Proof. In natural coordinates the Fisher flow is $\dot{\theta}_i = -\partial_{\mu_i}\mathcal{F}_I$, and $\partial_\mu D_{\mathrm{KL}}(p_\theta \| p_{\theta'}) = \theta - \theta'$, so $\dot{\theta}_i = c - \theta_i$, the rows summing to one. With $H(\mu) = \nabla^2\psi(\theta(\mu))$, the covariance of t, the pooled velocity is $\frac{1}{2}\sum_i H(\mu_i)(c - \theta(\mu_i))$. It is affine in c. Requiring it to be constant on each fiber $\{(\mu_1, 2\bar{\mu} - \mu_1)\}$ for every c in the open set Θ makes both $H(\mu_1) + H(2\bar{\mu} - \mu_1)$ and $K(\mu_1) + K(2\bar{\mu} - \mu_1)$ independent of μ_1, where $K(\mu) = H(\mu)\theta(\mu)$. The mean-parameter domain is open and convex. The first condition gives $DH(\mu_1) = DH(\mu_2)$ for all μ_1, μ_2, so H is affine. Since $D\theta = H^{-1}$, $DK(\mu)[v] = DH[v]\theta(\mu) + v$, and the second condition makes $DH[v]\theta(\mu)$ independent of μ; as $\theta(\mu)$ ranges over the open set Θ, $DH[v] = 0$ for every v. Hence $\nabla^2\psi$ is constant and ψ is quadratic. The Laplace transform of the law of t under the base measure is e^ψ, so by its uniqueness the laws of t

are Gaussian with covariance $\nabla^2\psi$; for $d = 1$ this is the constant-variance case of the classification of Morris [13]. □

For Bernoulli laws $H(p) = p(1-p)$. Two members with $p = (0.5, 0.5)$ and $p = (0.2, 0.8)$ have the same pool and the same rows, and with a single source of natural parameter 1 their pooled velocities are 0.25 and 0.16 (check (4)); no update of the pool reproduces both. The theorem concerns the pooled parent only. Another retention, such as the canonical parent of chapter 5 of the main manuscript for the routed interface, may still be exact for general laws; whether it compresses is open (open problem 8.1).

Remark 4.4 (What the parent forgets). The fibers of R are transient. By (iv) each member's deviation converges at the rates of $I + \rho$ to $(I + \rho)^{-1}\rho(o_i - \bar{o}_U)$, so on the attracting set each member's law is determined by its parent's mean and its own record. The parent forgets the members' records except their mean, and the transient part of their deviations; it forgets nothing that the members' own records do not restore. The fibers are inert in the sense used in chapter 11 of the main manuscript: no presented law, row or record depends on them. The interface on which R is exact does not exist before the handoff. While the members attend individually, their Gibbs rows, and the rows of every receiver attending to them, depend on each member's own mean, and the block mean is not autonomous. The handoff creates this operational equivalence; it is not present for every partition beforehand, which distinguishes the construction from the quotients supplied before membership in chapters 12 and 13 of the main manuscript. It is created by a declared rule, however, not by dynamics that are unchanged at the handoff.

Chapter 5

An instance

Take $N = 12$, $K = 1$, $\Lambda = \lambda = 6$, $T = 1$, homogeneous priors $\pi_s = 0.05$ and $\pi_p = 0.95/11$, evaluator mean $m_G = 0$, and $\rho = 0.005$. The records form four groups of three, $L_1 = (-3.2, -3.05, -2.9)$, $L_2 = (-2.1, -1.95, -1.8)$, $R_1 = -L_2$ and $R_2 = -L_1$, held by agents 0 to 2, 3 to 5, 6 to 8, and 9 to 11. The thresholds are $\delta_c = 0.05$, $\delta_w = 0.5$ and $\tau_p = 2$. Every cost is $\frac{\lambda}{2}$ times a squared difference of means, so two units are coherent when their means differ by less than 0.129 and walled when they differ by more than 0.408. The admitted class is

$$\mathcal{A} = \{y(0) \in \mathbb{R}^{12} : |y_i(0) - o_i| < 0.005 \text{ for all } i\}, \tag{5.1}$$

which is open. No pair is coherent at $t = 0$, since adjacent records differ by 0.15 and $\frac{\lambda}{2}(0.14)^2 = 0.0588 > \delta_c$. The members of a group disagree in their records, so a group becomes coherent only if the interaction contracts it; this is the device of chapter 13 of the main manuscript. The flows are those of theorem 4.1(ii) after the handoffs and of the per-agent Gibbs flow with individual rows before them.

Proposition 5.1 (The null flow never selects). From every state in $\mathcal{A}$ the null flow never selects a block, so its partition sequence is the initial partition alone.

Proof. Under Φ^0 each agent follows $\dot{y}_i = \beta^0_{ii}(y_i)(m_G - y_i) + \rho(o_i - y_i)$ with β^0_{ii} of eq. (1.2) and $D_{ii} = \frac{\lambda}{2}y_i^2$, so $\beta^0_{ii}(y) \le \frac{\pi_s}{1-\pi_s}e^{-3y^2}$. On $|y| \ge 1.795$ this is at most 3.3×10^{-6}, and the evaluator term is at most 1.1×10^{-5} in absolute value for $|y| \le 3.205$. At $y = o_i \pm 0.005$ the record term has magnitude 2.5×10^{-5} and points toward o_i, so the interval $[o_i - 0.005, o_i + 0.005]$ is forward invariant. Every pair of agents then stays at distance at least 0.14, with divergence at least $0.0588 > \delta_c$, so no pair is ever coherent and no candidate exists. □

Check (8) integrates the null flow to $t = 1000$: the agents end within 0.0012 of their records, and the smallest divergence between adjacent agents is 0.0675.

Remark 5.2 (Selections of the shared-label flow (numerical)). From every tested state in $\mathcal{A}$, fifteen in all (the fixed draw of checks (7) to (10), twelve uniform draws in check (12), and one draw each in checks (11) and (14)), the shared-label flow selects L_1, L_2, R_1 and R_2 between $t = 2.14$ and $t = 2.35$, and then $L_1 \cup L_2$ and $R_1 \cup R_2$ between $t = 6.95$ and $t = 8.05$. The selected blocks are the same for every state. Along the fixed-draw trajectory each group enters the candidate condition transversally, the slope of its coherence margin being at least 0.092 at entry, and over the rest of its persistence window the coherence margin, δ_c minus the largest edge of the group's minimum spanning tree, stays above 0.0046 while the wall margin stays above 0.50 (check (17)). The gap between the left parents after the first stage lies 0.35 below $d_u - 0.005$ (check (9) and corollary 5.4). By continuous dependence on initial data, which holds across the switching times of

the handoffs, the same blocks are selected on an open neighborhood of the fixed draw, provided the integration error stays below these margins. The integration is fourth-order Runge–Kutta with fixed steps between 0.005 and 0.05; it was not validated by interval arithmetic.

Proposition 5.3 (The second stage reduces to a gap equation). After the four first-stage handoffs write $m_1 = \bar{y}_{L_1}$, $m_2 = \bar{y}_{L_2}$, $d = m_2 - m_1$, and $s = \bar{o}_{L_2} - \bar{o}_{L_1} = 1.1$. While the left parent means lie in $[-3.3, -1.7]$ and the right ones in $[1.7, 3.3]$,

$$\dot{d} = F(d) + r(t), \qquad F(d) = \rho(s - d) - 2w_3(d)d, \qquad w_n(d) = \frac{ne^{-ad^2}}{n - 1 + ne^{-ad^2}}, \tag{5.2}$$

with $a = n\lambda/(2T) = 9$ for $n = 3$ and $|r(t)| \le 10^{-10}$.

Proof. By theorem 4.1(i) the row of L_1 has the evaluator with prior π_s and cost $9m_1^2$, its own parent with prior $2\pi_p$ and cost zero, L_2 with prior $3\pi_p$ and cost $9d^2$, and R_1, R_2 with prior $3\pi_p$ each and cost $9(m_1 - \bar{y}_R)^2$. On the stated ranges the evaluator weight relative to $2\pi_p$ is at most $\frac{\pi_s}{2\pi_p} e^{-26.0} < 1.5 \times 10^{-12}$, and the right-side weights are below e^{-100}. Dropping these weights from (iii) leaves $\dot{m}_1 = w_3(d)(m_2 - m_1) + \rho(\bar{o}_{L_1} - m_1)$, the prior π_p cancelling from w_3, and the same for m_2 with the roles exchanged; the dropped terms contribute less than 10^{-10} to $\dot{d}$. Subtracting the two equations gives eq. (5.2). The stated ranges are forward invariant up to these terms, since each parent is pulled toward the other left parent and toward its own record mean, both inside the range. □

The function F has exactly three zeros in $(0, \infty)$, $d_m = 0.00456$, $d_u = 0.96304$ and $d_s = 1.08312$, with $F'(d_m) = -1.205$, $F'(d_u) = 0.0062$ and $F'(d_s) = -0.0034$ (check (9)). For $d > s$ both terms of F are negative, and the zeros below s were located by sign changes on a grid of spacing 3×10^{-6} and refined by bisection. The merged state d_m and the separated state d_s are stable, and the middle root d_u separates their basins. The separated state is an equilibrium held by the records: without them, $\rho = 0$, the only zero is $d = 0$.

Corollary 5.4 (Counterfactual handoffs). Consider post-handoff states in which L_1, L_2, R_1, R_2 are the selected blocks and the left parent means lie in $[-3.3, -1.7]$ with gap d_0. If $d_0 < d_u - 0.005$, the union $L_1 \cup L_2$ is selected. If $d_0 > d_u + 0.005$, it is never selected, and $d(t) \to d_s$.

Proof. By check (16), $F < -3.2 \times 10^{-5}$ on $[0.129, d_u - 0.005]$, $F > 1.6 \times 10^{-5}$ on $[d_u + 0.005, d_s - 0.005]$, and $F < -1.7 \times 10^{-5}$ on $[d_s + 0.005, 1.6]$, while $|r| \le 10^{-10}$. In the first case d decreases through 0.129 in finite time and then stays below it near d_m, where F is positive below d_m and negative above. The units L_1 and L_2 are then coherent and walled from the right side, so the union is selected after the window τ_p. In the second case the same signs keep d above $d_u + 0.005$ for all time and drive it to within 10^{-7} of d_s, so L_1 and L_2 are never coherent. □

The three handoffs whose gap was recorded have gaps between 0.59 and 0.63 at $t = 2.6$, after all four first-stage handoffs (checks (9), (10) and (14)), and so lie in the first case; all fifteen tested states unite. The counterfactual handoffs of check (10) shift L_1 and L_2 apart after the first stage. Gaps of 0.595, 0.795 and 0.895 unite, at $t = 7.6$, 18.4 and 61.6; gaps of 1.095 and 1.395 converge to the separated state 1.0831. Every one of these states is walled, since its gap exceeds 0.408, so all of them lie in the inherited class.

Remark 5.5 (Controls and depth (numerical)). The control without feedback unites from every shifted handoff of check (10), gaps 0.45 to 1.25, at times between 5.9 and 11.4. At synchrony its gap equation is eq. (5.2) with $a = \lambda/(2T) = 3$, which has only the merged zero (check (9)), as proposition 3.4 predicts; the tempered label produces the same events as the control (check (8)). With record gap $s = 1.2$ or 1.3 the first stage is the same, but the handoff gap, 0.996 or 1.226, exceeds the middle root, 0.917 or 0.893, and the parents converge to the separated zero, 1.198 or 1.300 (check (14)). The hierarchy stops at depth one there, while the control unites (check (11)). With $s = 1.0$ the members of L_1 and L_2 become coherent with one another before either group is walled, and a single block of six forms at the first stage. With $s = 1.1$ the middle root is 0.963 for $n = 3$, 0.771 for $n = 4$ and 0.593 for $n = 6$, and there is no separated zero for $n = 2$ (check (13)). For large n, replacing w_n by $\frac{n}{n-1}e^{-ad^2}$ in $F = 0$ gives

$$d_u^2 \approx \frac{2T}{n\lambda} \log \frac{2nd_u}{(n-1)\rho(s-d_u)}, \tag{5.3}$$

which reproduces 0.593 at $n = 6$. The capture distance falls roughly as $n^{-1/2}$: larger blocks unite only with nearer partners. This suggests that a hierarchy whose higher-level clusters are farther apart stops at finite depth under exact label sharing; we have not proved it.

Chapter 6

The formation standard

Each clause of the definition of an inherited recursive formation in chapter 5 of the main manuscript is checked for the instance of chapter 5.

(F1) The admitted class eq. (5.1) is a nonempty open subset of $\mathbb{R}^{12}$, and its states are initially unpartitioned.

(F2) The selector of definition 1.1 is declared, with persistence window τ_p, and each handoff is a measurable function of accessible history. The selected blocks are nontrivial and are never dissolved. After a handoff each difference between two members' deviations moves, by theorem 4.1(iv), along a segment from its value at the handoff to its record-determined limit, which is at most 0.0015 in absolute value in the first stage and 0.007 in the unions, far below the coherence radius 0.129.

(F3) By theorem 4.1 the retention to unit means satisfies (D), (B) for the routed protocols, in which a receiver outside a block attends to its parent, and (C). The protocol class is fixed by definition 3.1 before the retention. Condition (G) is not declared.

(F4) The first-stage retention forgets the deviations of each block, a subspace of dimension $n_U - 1$, and the second-stage retention forgets the relative coordinate $\bar{y}_{L_1} - \bar{y}_{L_2}$. Configurations in one fiber stay distinct. A child-level probe, a receiver attending to one member through an identity channel as before the handoff, reads y_i and distinguishes them; at the second stage a receiver attending to A_{L_1}, as before the union, does the same. The fibers are transient and inert (remark 4.4).

(F5) By proposition 5.1 the null flow never selects, while the shared-label flow selects six blocks (remark 5.2), so $P \neq P^0$ on the tested states and, by the margins of remark 5.2, on an open neighborhood of the fixed draw. The strong form fails: the final partition is $\{L_1 \cup L_2, R_1 \cup R_2\}$ on every tested state, and a constant is a function of any statistic. Amplification, as defined, compares event ratios along Φ and Φ^0, and Φ^0 has no events, so it is not assessed.

(F6) The parents A_{L_1} and A_{L_2} select their union under the inherited parent flow (iii), with no reset of their source data and no new parameter, which is (F6a). By corollary 5.4 the union is not selected from counterfactual handoffs in the inherited class with gap above $d_u + 0.005$, which is (F6b). The dependence of the second stage on the first is a selection between two basins by the state that the first stage produces; the control without feedback shows no such dependence on the tested handoffs.

Clause	Gaussian location instance	General laws
(F1)	holds	holds for the same class
(F2)	holds (declared thresholds)	holds (declared thresholds)
(F3)	holds exactly, theorem 4.1	fails for the pooled parent, theorem 4.3; open
(F4)	holds; fibers transient and inert	open
(F5)	holds (open set by numerical margins); strong form fails	not established
(F6a)	holds (open set by numerical margins)	not established
(F6b)	holds, corollary 5.4	not established

In the instance the construction satisfies the six clauses as they are stated, the open-set parts of (F5) and (F6a) resting on the numerical margins of remark 5.2. With that proviso it is an inherited recursive formation in that instance, subject to qualifications that also mark where it falls short of the preferred form of the open problem on recursive formation in chapter 5 of the main manuscript, which asks for a derived selector and a derived interface. The selector thresholds are declared; chapter 7 replaces them by a comparison of free energies. The label-sharing handoff is declared, and the compression interface is derived from it; the operational equivalence on which the parent is exact appears at the handoff and is absent before it, but it is created by the declared change of model, not by unchanged dynamics. The compressed coordinates are transient, and on the attracting set the members are determined by their parent and their own records. Exactness is a property of the Gaussian location family (theorem 4.3). The first-stage events and the unions from the actual handoffs are established by fixed-step numerical integration, with explicit margins but without validated arithmetic, while the counterfactual statement rests on eq. (5.2), whose neglected terms are bounded explicitly and whose zeros and sign margins were computed numerically. The parameter window is narrow: with record gap 1.0 the first stage already fuses the two groups, and with gaps 1.2 and 1.3 the hierarchy stops at depth one.

For general laws the selector, the shared row and its n-fold sharpening are unchanged (proposition 3.2). The pooled parent is not exactly inherited outside the Gaussian location family, so (F3) is open there, (F4) depends on it, and (F5) and (F6) have not been established beyond the instance.

Chapter 7

A derived selector

The selector of definition 1.1 declares two thresholds and a window. This chapter replaces the thresholds by a comparison of free energies: a block is selected when the models that the handoff would create have a lower free energy than the current ones. The handoff rule definition 3.1, the flows and the instance are unchanged, and the window τ_p remains declared.

Definition 7.1 (Structure free energy and the derived selector). Let $\mathcal{U}$ be the current units. A coarsening of $\mathcal{U}$ is a partition $\mathcal{Q}$ of $\mathcal{V}$ each of whose blocks is a union of units of $\mathcal{U}$. For a block $U \in \mathcal{Q}$ with n_U agents, let $\mathcal{F}_U(\mathcal{Q})$ be the minimum over rows of the block free energy eq. (3.1) of U under one shared label, whose sources are the evaluator, the parent A_U when $n_U \geq 2$, and the other blocks of $\mathcal{Q}$, each presenting its law as in definition 3.1(b), with the priors of definition 3.1(a); for a single agent it is the minimum of eq. (1.1) over the row. The structure free energy of $\mathcal{Q}$ is

$$\mathcal{F}(\mathcal{Q}) = \sum_{U \in \mathcal{Q}} \mathcal{F}_U(\mathcal{Q}), \tag{7.1}$$

and $\mathcal{F}(\mathcal{U})$ is the sum of the current free energies of the units. At each time let $\mathcal{Q}^*(t)$ be a minimizer of $\mathcal{F}$ over the coarsenings of $\mathcal{U}(t)$ at the current laws. A block of $\mathcal{Q}^*(t)$ that is not a unit is a candidate, and a handoff selects it once it has been a candidate without interruption for a time τ_p. Several candidates may be selected at one time, and a selected block is never dissolved.

The record terms of eq. (7.1) are the same for every coarsening at fixed laws, so the comparison reads only the laws and the source priors. The sum runs over all blocks of $\mathcal{Q}$, because merging some units also changes the sources of the others. The selector is a measurable function of the history of the laws, through the candidate clocks, as (F2) requires; the clocks belong to the state of the process, as those of definition 1.1 do. By the definition of the null history in chapter 5 of the main manuscript, the same selector is applied along Φ^0: it evaluates eq. (7.1) with the actual costs at the laws of the null flow. At $T = 1$ each $\mathcal{F}_U$ is the variational free energy of a normalized model of the members' contents and records with the laws of the other blocks as data, so minimizing eq. (7.1) compares evidence bounds under a uniform prior over coarsenings. This is structure learning in the sense of Czégel et al. [3], and greedy merging by such comparisons is the scheme of Bayesian hierarchical clustering [7].

Proposition 7.2 (The structure free energy reads spreads and pools). Under the hypotheses of definition 3.1, let every parent present the linear pool of its members. For every coarsening $\mathcal{Q}$ and every block $U \in \mathcal{Q}$,

$$\mathcal{F}_U(\mathcal{Q}) = \sum_{i \in U} D_{\mathrm{KL}}(q_i \| \bar{q}_U) - T \log\Big(\pi_{UA} + \sum_{\nu} \pi_{U\nu} e^{-n_U D_{\mathrm{KL}}(\bar{q}_U \| \nu_\nu)/T}\Big) + \sum_{i \in U} \int -\log \ell_i(o_i \mid k) q_i(dk), \tag{7.2}$$

where ν runs over the evaluator and the other blocks of $\mathcal{Q}$, and $\pi_{UA} = 0$ when $n_U = 1$. If the compared laws lie in a regular exponential family E and every parent presents the moment projection P_U of its pool, the same holds with $\sum_{i\in U} D_{\mathrm{KL}}(q_i\|P_U)$ as the first term and $D_{\mathrm{KL}}(P_U\|\nu_\nu)$ in the exponent. In the Gaussian location family of theorem 4.1, with homogeneous priors π_s and π_p and evaluator mean m_G, the structure free energy minus the record terms is

$$\frac{1}{2}\sum_{U\in\mathcal{Q}}\sum_{i\in U}\|y_i-\bar{y}_U\|^2_\Lambda - T\sum_{U\in\mathcal{Q}}\log\Big((n_U-1)\pi_p+\pi_s e^{-\frac{n_U}{2T}\|\bar{y}_U-m_G\|^2_\Lambda}+\sum_{V\neq U} n_V\pi_p e^{-\frac{n_U}{2T}\|\bar{y}_U-\bar{y}_V\|^2_\Lambda}\Big). \quad (7.3)$$

Proof. By eq. (3.2) the summed cost of a source ν for the members of U is $\sum_{i\in U} D_{\mathrm{KL}}(q_i\|\bar{q}_U) + n_U D_{\mathrm{KL}}(\bar{q}_U\|\nu_\nu)$, and that of the parent is the first term alone. The minimum over rows of $\sum_\nu \beta_\nu c_\nu + TD_{\mathrm{KL}}(\beta\|\pi)$ is $-T\log\sum_\nu \pi_\nu e^{-c_\nu/T}$, the Gibbs variational principle. Factoring $e^{-\sum_i D_{\mathrm{KL}}(q_i\|\bar{q}_U)/T}$ out of the sum gives eq. (7.2). For moment projections, eq. (3.4) gives $D_{\mathrm{KL}}(\bar{q}_U\|\nu_\nu) = D_{\mathrm{KL}}(\bar{q}_U\|P_U) + D_{\mathrm{KL}}(P_U\|\nu_\nu)$ for every $\nu_\nu \in$ E, and eq. (3.2) with P_U in place of ν gives $\sum_i D_{\mathrm{KL}}(q_i\|\bar{q}_U) + n_U D_{\mathrm{KL}}(\bar{q}_U\|P_U) = \sum_i D_{\mathrm{KL}}(q_i\|P_U)$, which is also the parent's cost. In the Gaussian location family $D_{\mathrm{KL}}(\mathcal{N}(y,\Lambda^{-1})\|\mathcal{N}(m,\Lambda^{-1})) = \frac{1}{2}\|y-m\|^2_\Lambda$, and by definition 3.1(a) the parent of U receives the mass $(n_U-1)\pi_p$ that the members gave to one another, a block V the mass $n_V\pi_p$, and the evaluator π_s. □

Check (18) confirms both general forms on a six-point alphabet, and check (20) confirms eq. (7.3) against explicit Gibbs rows to 6×10^{-14}. The first term of eq. (7.2) is n_U times the generalized Jensen–Shannon divergence of the members with equal weights [9]; in the Gaussian location family it is the within-block sum of squares in the Fisher metric, the objective of centroid clustering [10]. The second term is the free energy, at temperature T/n_U, of a single receiver that holds the pool and has one additional source at cost zero, the block's own parent, carrying the mass that the members gave to one another. For a block far from every other source the second term is close to $-T\log((n_U-1)\pi_p)$, and eq. (7.3) becomes a centroid criterion with a penalty for each block, of the kind that Dirichlet-process mixtures give in the limit of small variance [8]; here the penalty depends on the block size and is fixed by the source prior instead of being declared.

Proposition 7.3 (Mutual attention favors a merge). Let $i \neq j$ be single-agent units that compare each other through identity channels, with Gibbs rows β_i, β_j, mutual priors π_{ij}, π_{ji} and costs D_{ij}, D_{ji}, and let $\mathcal{Q}$ merge them, so that their parent has prior $\pi_A = (\pi_{ij}+\pi_{ji})/2$ by definition 3.1(a). Let the parent present the linear pool, or the moment projection onto a family that contains q_i and q_j. Then

$$\mathcal{F}_i + \mathcal{F}_j - \mathcal{F}_{\{i,j\}}(\mathcal{Q}) \geq T\log\frac{\beta_{ij}\beta_{ji}\pi_A}{\pi_{ij}\pi_{ji}} + \max(D_{ij}, D_{ji}). \quad (7.4)$$

Let an agent v outside the pair compare i and j through one Markov channel, with costs D_{vi}, D_{vj} and current attention $\beta_{v\{ij\}} = \beta_{vi} + \beta_{vj}$ to the pair, and let it attend to the parent through the same channel with prior $\pi_{vi} + \pi_{vj}$. If the parent is the linear pool, or in the Gaussian location family its moment projection, then

$$\mathcal{F}_v(\mathcal{Q}) - \mathcal{F}_v \leq -T\log\Big(1-\beta_{v\{ij\}}\big(1-e^{-|D_{vi}-D_{vj}|/(2T)}\big)\Big). \quad (7.5)$$

Proof. Write Z_i for the normalizer of the row of i. Since $\beta_{ij} = \pi_{ij}e^{-D_{ij}/T}/Z_i$, we have $\mathcal{F}_i = -T\log Z_i = -T\log\pi_{ij} + D_{ij} + T\log\beta_{ij}$, and likewise for j. The normalizer of the pair's shared row contains the term of its parent, so $\mathcal{F}_{\{i,j\}}(\mathcal{Q}) \le -T\log\pi_A + D_{iA} + D_{jA}$, where D_{lA} is the cost of member l for the parent. In both cases the parent's law minimizes the summed cost $D_{\mathrm{KL}}(q_i\|p) + D_{\mathrm{KL}}(q_j\|p)$ over the laws p it may present, by eq. (3.2) and, for the projection, eq. (3.4). The laws q_j and q_i are among them, so $D_{iA} + D_{jA} \le \min(D_{ij}, D_{ji})$. Adding the three relations gives eq. (7.4), because $D_{ij} + D_{ji} - \min(D_{ij}, D_{ji}) = \max(D_{ij}, D_{ji})$.

For v, only the terms of i and j in its normalizer Z_v change. Before the merge they sum to $b \le (\pi_{vi} + \pi_{vj})e^{-\min(D_{vi}, D_{vj})/T}$. After it the parent's term is $(\pi_{vi} + \pi_{vj})e^{-D_{vA}/T}$, and $D_{vA} \le \frac{1}{2}(D_{vi} + D_{vj})$: for the pool by convexity of D_{KL} in its second argument, after the channel, which maps the pool to the pool of the channelled laws; in the Gaussian location family by convexity of the squared norm. Since $\frac{1}{2}(D_{vi} + D_{vj}) - \min(D_{vi}, D_{vj}) = \frac{1}{2}|D_{vi} - D_{vj}|$, the new term is at least $be^{-|D_{vi} - D_{vj}|/(2T)}$, so the new normalizer is at least $Z_v - b\big(1 - e^{-|D_{vi} - D_{vj}|/(2T)}\big)$, and $b = \beta_{v\{ij\}}Z_v$. □

The first term of eq. (7.4) is a saving of labels, the Occam factor of Bayesian model comparison [11]: two agents that attend to each other pay for two independent choices of source in their own models and for one in the shared model, whose parent collects the mass they gave to each other. The second term is a double use of the members' laws. The parent presents the law closest to both members in summed divergence, so each member is compared with a law that contains itself, and the bound grows with the divergence between the members as long as they attend to each other. A cavity parent, which presents to each member the pool of the other members, removes the double use: for a pair its cost is $D_{ij} + D_{ji}$, and the bound keeps only its first term. It also makes the cost of the own parent differ from the cost common to all sources by a multiple of the spread, so the shared row depends on the spread and theorem 4.1 fails. The saving of labels remains with the cavity parent. Check (19) confirms both bounds on 2000 random draws on a five-point alphabet, with smallest slacks 0.088 and 0.0011.

Corollary 7.4 (The derived selector reads initial attention). Let the peer priors be homogeneous, $\pi_{ij} = \pi_p$. At a state where two agents satisfy

$$T\log\frac{\beta_{ij}\beta_{ji}}{\pi_p} + \max(D_{ij}, D_{ji}) > -T\sum_{v\neq i,j}\log\Big(1 - \beta_{v\{ij\}}\big(1 - e^{-|D_{vi} - D_{vj}|/(2T)}\big)\Big), \tag{7.6}$$

the partition into single agents does not minimize eq. (7.1), so the derived selector has a candidate there. In particular, if such a pair exists at a state of the admitted class, the selector has a candidate at time zero along both Φ and Φ^0.

Proof. By proposition 7.3, merging the pair lowers $\mathcal{F}$ by at least the left side minus the right side of eq. (7.6). The two flows start from the same state, and the selector is the same function of the state. □

With the declared selector an admitted class could be chosen on which the null flow never selects (proposition 5.1), because the coherence radius 0.129 lies below the spacing of the records. The derived selector has no radius to choose. It groups agents that attend to one another, and the records alone make agents attend to one another when nothing else lies closer. A first stage is then

interaction-necessary only if the interaction changes the minimizer of eq. (7.1) within the window. At the fixed draw of the instance, $\beta_{01}\beta_{10} = 0.254$ and $\beta_{12}\beta_{21} = 0.229$ exceed $\pi_p = 0.086$; for the pair $\{0, 1\}$ the members gain 1.737 against the bound 1.145, and the outsiders' free energies change by -0.0035. Over all $4\,213\,597$ partitions of the twelve agents, the minimizer of eq. (7.1) at the fixed draw is the four groups, with margin 1.715 to the next (check (21)).

Remark 7.5 (Derived selections in the instance (numerical)). Take the instance of chapter 5 with the derived selector in place of definition 1.1. Since $K = 1$, the minimizer was computed over the interval coarsenings of the units ordered by their means; at the fixed draw it agrees with the minimizer over all partitions, and at five later states with the minimizer over the $41\,209$ partitions that do not mix the two sides (checks (21) and (22)). Along the flow with individual rows from the fixed draw, the minimizer is the four groups from $t = 0$, with margin at least 0.86 on $[0, 1.3]$. At $t = 1.75$ the union $L_1 \cup L_2$ replaces L_1 and L_2, and from $t = 1.77$ the minimizer is $\{L_1 \cup L_2, R_1 \cup R_2\}$. The gap between the means of L_1 and L_2 falls below $d_u - 0.005$ at $t = 0.96$. On the box $\prod_i [o_i - 0.005, o_i + 0.005]$, which is forward invariant under Φ^0 by the proof of proposition 5.1, also after handoffs because a shared null row gives the evaluator a smaller weight than an individual one, the minimizer is the four groups at each of 200 sampled states, with margin at least 1.82. The selector is evaluated after every integration step of length 0.05, so a candidate present from the start is selected at $\tau_p + 0.05$.

Three regimes follow (check (23)). With $\tau_p = 0.5$ both flows select the four groups at $t = 0.55$, before the left gap has fallen below $d_u - 0.005$; the shared-label parents stay separated, with left gap 1.056 at $t = 300$, and the partition sequences of Φ and Φ^0 coincide. With $\tau_p = 1.25$ both flows select the four groups at $t = 1.30$, the shared-label flow then selects $L_1 \cup L_2$ and $R_1 \cup R_2$ at $t = 40.3$ and 44.15, and the null flow selects nothing more; over twelve further states of $\mathcal{A}$ the partition sequences are the same, with unions between $t = 36.5$ and 60.9 (check (24)). With $\tau_p = 2$, the window of chapter 5, the minimizer changes before the window ends, the shared-label flow selects $L_1 \cup L_2$ and $R_1 \cup R_2$ as first-stage blocks at $t = 3.75$ and 3.80, the null flow selects the four groups at $t = 2.05$, and there is no second stage. The switch times of the fixed draw suggest that two stages form whenever the handoff falls in $[0.96, 1.75)$, that is for $0.91 \le \tau_p < 1.70$; only $\tau_p = 1.25$ was run in that range. The control without feedback, with $\tau_p = 1.25$, selects the four groups at $t = 1.30$ and their unions at $t = 3.0$ and 3.05.

Proposition 7.6 (The derived union criterion is a radius). After the four first-stage handoffs, let the parent means lie in the ranges of proposition 5.3 and write $d = m_2 - m_1$. The minimizer of eq. (7.1) over the coarsenings of the four units contains $L_1 \cup L_2$ when $\Gamma_3(d) > 10^{-10}$ and does not when $\Gamma_3(d) < -10^{-10}$, where, for two parents of n members each,

$$\Gamma_n(d) = \log \frac{2n-1}{\pi_p} - 2\log\bigl(n - 1 + ne^{-n\lambda d^2/(2T)}\bigr) - \frac{n\lambda}{4T} d^2. \tag{7.7}$$

If $(2n-1)\pi_p < 1$, then Γ_n has exactly one zero $d_*(n)$ in $(0, \infty)$, is positive below it and negative above it, and is decreasing above it. In the instance, with $T = 1$ and $\lambda = 6$, $d_*(3) = 0.7685$.

Proof. For the union, $\sum_{i \in L_1 \cup L_2} (y_i - \bar{y})^2 = W_{L_1} + W_{L_2} + \frac{n_1 n_2}{n_1 + n_2} d^2$, the decomposition of a sum of squares about the pooled mean, so the spreads of L_1 and L_2 cancel between the two coarsenings, whatever their values, and the union adds $\frac{\lambda}{2} \cdot \frac{3}{2} d^2$. By eq. (7.3) each separate parent contributes

$-\log(2\pi_p + 3\pi_p e^{-9d^2})$ and the union $-\log(5\pi_p)$, once the evaluator and the right units are dropped from the normalizers. As in the proof of proposition 5.3, the evaluator's weight relative to $2\pi_p$ is below 1.5×10^{-12} for a left parent and smaller still for the union, and every weight between the two sides is below e^{-100}, which also bounds the change in the right units' free energies. Hence $\mathcal{F}(\text{separate}) - \mathcal{F}(\text{union}) = \Gamma_3(d) + e$ with $|e| < 10^{-10}$. Every other coarsening joins a left and a right unit, whose means differ by at least 3.4, so its spread term is at least $\frac{\lambda}{2} \cdot \frac{3}{2}(3.4)^2 > 50$; each attention term of a block of at least three members lies in $[0, -\log(2\pi_p)]$, so the attention terms of two coarsenings of the four units differ by less than 8, and such a coarsening has a larger structure free energy than the four separate units. The choices on the two sides are independent up to the same weights.

For the zeros put $u = d^2$ and $a = n\lambda/(2T)$. The function $h(u) = \log(n - 1 + ne^{-au})$ has $h''(u) = a^2 n(n-1)e^{-au}/(n-1+ne^{-au})^2 > 0$, so Γ_n is strictly concave in u. It has the value $-\log((2n-1)\pi_p) > 0$ at $u = 0$ and tends to $-\infty$. A strictly concave function with these properties has exactly one zero, is positive before it, negative after it, and decreasing after it, because its maximum lies before the zero. □

The radius lies between the merged root and the middle root of the gap equation, $d_m = 0.00456 < d_* = 0.7685 < d_u - 0.005 = 0.958$. For other block sizes $d_*(2) = 1.087$, $d_*(4) = 0.600$ and $d_*(6) = 0.410$; the value $\Gamma_n(0) = -\log((2n-1)\pi_p)$, the saving of labels when two synchronized parents merge, shrinks as the parents' share of the prior grows, down to $-\log(1 - \pi_s) = 0.051$ when the union holds all agents. Check (25) confirms the reduction to within 2×10^{-12} over 72 configurations in the stated ranges.

Corollary 7.7 (Counterfactual handoffs under the derived selector). Consider post-handoff states as in corollary 5.4, with gap d_0 and no running candidate clock. If $d_0 < d_u - 0.005$, the union $L_1 \cup L_2$ is selected. If $d_0 > d_u + 0.005$, the gap stays in $[d_u + 0.005, 1.6]$ for all time and the union is never selected.

Proof. The signs of F in eq. (5.2) on the three intervals of check (16), sampled as in corollary 5.4, together with $|r| \le 10^{-10}$, make d decrease while it lies in $[0.129, d_u - 0.005]$ and make $[d_u + 0.005, 1.6]$ forward invariant. In the first case d falls below d_*, which lies in that interval, in finite time and stays below it, so by proposition 7.6 the union is a block of the minimizer from then on and is selected after τ_p. In the second case $d \ge d_u + 0.005 > d_*$ for all time, so $\Gamma_3(d) < \Gamma_3(d_u) < 0$ and the union is never a candidate. □

Check (26) shifts L_1 and L_2 apart after the handoffs at $\tau_p = 1.25$. Handoff gaps of 0.59, 0.79, 0.89 and 0.94 unite, at $t = 2.65$, 5.3, 44.15 and 155.6; gaps of 0.99, 1.09 and 1.39 do not unite by $t = 2000$ and end between 1.0829 and 1.0833, near d_s. The control without feedback unites from every tested gap, 0.879 to 1.679, at times between 3.05 and 62.0.

Remark 7.8 (The formation standard under the derived selector). For handoffs in $[0.96, 1.75)$, tested at $\tau_p = 1.25$, the clauses read as follows. (F1) is unchanged. (F2) holds with a derived comparison; the window τ_p, the uniform prior over coarsenings and the handoff rule are declared. (F3) and (F4) are as in chapter 6, with the qualifications stated there, because the handoff rule is unchanged and theorem 4.1 holds between handoffs after any sequence of shared-label handoffs; for either selector, a complete retained parent must also carry the candidate clocks. For (F5), the

null history on the tested states is the four groups at $\tau_p + 0.05$ and nothing after: on the invariant box the left gap stays within 0.01 of 1.1, so $\Gamma_3 < -2.5$ there by proposition 7.6, and likewise on the right. The shared-label flow selects the unions, so $P \neq P^0$ on the tested states and, by the margins of remark 7.5, on an open neighborhood of the fixed draw, under the same proviso on integration error as in remark 5.2. The first stage is not interaction-necessary: the null flow selects the same blocks at the same time, and they minimize eq. (7.1) at the initial state. The difference lies in the second stage alone, and the strong form of (F5) fails. (F6a) holds on the tested states, and (F6b) holds by corollary 7.7. With $\tau_p = 2$ the construction is a single-stage selection with exact inheritance whose stage is interaction-necessary, since Φ selects the two unions and Φ^0 the four groups. With $\tau_p = 0.5$ the two flows have the same partition sequence and (F5) fails.

Clause	Declared selector	Derived, handoff in $[0.96, 1.75)$	Derived, $\tau_p = 2$
(F1)	holds	holds	holds
(F2)	declared thresholds and window	derived comparison, declared window	derived comparison, declared window
(F3)	holds exactly, theorem 4.1	holds exactly, theorem 4.1	holds exactly, theorem 4.1
(F4)	holds; fibers transient and inert	holds; fibers transient and inert	holds; fibers transient and inert
(F5)	holds at the first stage (numerical margins)	holds at the second stage only (numerical margins)	holds at the only stage (numerical)
(F6a)	holds (numerical margins)	holds (numerical)	fails: no second stage
(F6b)	holds, sampled sign margins	holds, sampled sign margins	not applicable

The derived selector removes the two thresholds but not the window, which takes over their role. Two times matter: the time at which the interaction has brought the groups within the middle root of the gap equation, and the later time at which the minimizer itself moves to the unions. A first handoff before the first time stops the hierarchy at depth one, a handoff between them gives two stages, and a window that outlasts the second gives a single coarser stage. No window gives interaction necessity at both stages in this instance: until $t = 1.75$ the minimizer along both flows is the four groups, and after that the minimizer of the shared-label flow is already the pair of unions. A prior over coarsenings that offsets the saving of labels, or a cavity parent at the cost of exact inheritance, would change the radius and the regimes (open problem 8.3).

Chapter 8

Prior work and open problems

The ingredients have precedents. Inferred group membership that feeds back on individual updating appears in the meta-contrast model of Salzarulo [17], where categories are recomputed from the opinion context and agents move toward prototypes. It also appears in the latent-group model of Gershman et al. [6], where an observer infers group structure with shared choice parameters and the posterior determines social influence. Replacing locally coherent groups by representatives that act at the next stage is the hierarchical consensus scheme of Perony et al. [16], with prescribed stages and no feedback to the members. Martins [12] derived bounded confidence from Bayesian updating, with stubbornness from shrinking individual uncertainty. Normalized state-dependent weights are the setting of Motsch and Tadmor [14], and anchoring of opinions to fixed data is the model of Friedkin and Johnsen [4]. Exact block-mean dynamics of linear consensus over an external equitable partition is due to O'Clery et al. [15], and Schaub et al. [18] used these partitions for cluster synchronization in nonlinear networks. Multilevel selection has been written as Bayesian inference, with the formation of new levels as structure learning [3]. A literature search on 2026-10-23, a research run of the Exa agent followed by checking each cited entry against its DOI record or publisher page, found no work that combines a shared source label for dynamically formed groups, exact block-mean closure for state-dependent rows away from synchrony, and merging that depends on the state at formation. The search was bounded and does not establish priority.

Open problem 8.1 (General laws). Find an exactly inherited compressive parent for shared-label blocks outside the Gaussian location family. Candidates are a retention other than the pool, such as the canonical parent of chapter 5 of the main manuscript for the routed interface, or an approximate inheritance whose defect is controlled by the spread inside the block. Alternatively, show that no exact compressive parent exists for Bernoulli blocks with Gibbs rows.

Open problem 8.2 (A derived persistence criterion). The derived selector of chapter 7 keeps the window τ_p, and the window decides whether the hierarchy stops at depth one, forms two stages, or forms one coarser stage (remark 7.8). Replace it by a criterion read from the agents' models, for example accumulated evidence for the shared label in the sense of a sequential test [19], or the message counts of the registered-agreement model of chapter 3 of the main manuscript, and decide whether the two-stage regime survives.

Open problem 8.3 (Interaction necessity at both stages). Under the derived selector the first stage reads the attention present in the initial state (proposition 7.3 and corollary 7.4). Decide whether a prior over coarsenings that offsets the saving of labels, or a cavity parent together with an exact replacement for theorem 4.1, gives an admitted class on which the null flow does not select the first-stage blocks while the second stage still depends on the first, and whether some instance gives interaction necessity at both stages without either change.

Open problem 8.4 (Persistent compression). The coordinates discarded here are transient. Combine the shared label with the code-invisible sectors of Items 1 to 3, or with the flatness condition of remark 3.5, so that the parent forgets coordinates that persist and that an admitted child-level probe still reads.

Open problem 8.5 (The reciprocal potential). In the per-agent model a source feels no force from the receivers that attend to it. Under the reciprocal potential a parent that presents a pool receives source-role forces from every receiver, and Item 2 found that disagreement makes sources less certain there. Determine whether this loss of certainty offsets the n-fold sharpening of the shared row, and whether exact inheritance survives the source-role terms. This is the question about certainty raised in Item 3.

Open problem 8.6 (Depth). By eq. (5.3) the capture distance falls roughly as $n^{-1/2}$. Determine the depth reached from random records, and whether partial pooling, with the members' costs added with weight n^{p-1} for some $0 < p < 1$, allows any finite depth while keeping counterfactual dependence at every stage.

Open problem 8.7 (Validated numerics). Certify the first-stage events and the unions from the actual handoffs on an explicit open subset of $\mathcal{A}$ by interval integration, and with it the open-set form of (F5) and (F6a).

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